



**FACULTY OF COMPUTING**

**SEMESTER I, SESION 2024/2025**

**BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)**

**SECR1213 NETWORK COMMUNICATIONS - SECTION 01**

**Task 6b: MAKING AN INDIVIDUAL REPORT**

**NETWORK DESIGN FOR FACULTY OF COMPUTING BLOCK N28B  
by DATA VOYAGERS**

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| <b>LING YU QIAN</b> | <b>A23CS0301</b> |
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**LECTURER'S NAME : DR MUHAMMAD ZAFRAN BIN MUHAMMAD ZALY SHAH**

**SUBMISSION DATE : 25 JANUARY 2025**

**Individual video: <https://youtu.be/myWee4l9Uow>**

Our group name is DATA VOYAGERS, consisting of Chin Pei Wen, Koo Xuan, Ling Yu Qian, and Tan Zhao Hong

### **Detail your contribution to the group and the project work**

My contributions to the group and project work encompassed several key tasks. For Task 1, I designed the first-floor floor plan, ensuring accurate scaling, logical room layouts, and efficient space utilization to accommodate network components and workstations. In Task 2, I conducted thorough research and actively participated in discussions to analyze user requirements and evaluate project feasibility. For Task 3, I compared various devices based on their specifications and costs, aiding the group in selecting options that met both technical needs and budget constraints. In Task 4, I planned the network arrangement for the cable connection, determining the optimal placement of switches, cables, and ports to ensure seamless connectivity and scalability. For Task 5, I engaged in discussions and contributed to the subnetting plan, focusing on efficient IP allocation and resolving potential conflicts. Finally, in Task 6, our team collaboratively wrote a group report reflection, where I contributed to summarizing our efforts and discussing our overall project experience. These contributions reflect my active involvement in creating a well-structured and effective network solution.

### **Detail the other members contribution to the group and the project work**

Chin Pei Wen (A23CS0065)

Chin designed the ground floor plan (Task 1) and conducted an interview with the lecturer to assess project feasibility (Task 2). They researched and compared network devices to support selection (Task 3), drew network diagrams for work areas (Task 4), and contributed to subnetting and IP assignment (Task 5). Koo Xuan (A23CS0300) Koo designed the first-floor plan (Task 1) and conducted research on scalability, security, and cost-effectiveness (Task 2). They helped with device selection and budgeting (Task 3), determined device placement on the floor plan (Task 4), and contributed to subnetting discussions (Task 5). Tan Zhao Hong (A23CS0188) Tan wrote descriptions for the floor plans (Task 1), researched and answered specific questions (Task 2), compared and selected network devices (Task 3), created cabling diagrams (Task 4), and handled IP assignment for work areas (Task 5). The project involved designing floor plans, selecting and arranging network devices, creating network and cabling diagrams, and implementing subnetting and IP assignments to build a scalable and efficient network solution.

### **Explain how you work as a group.**

Before starting any task, we begin by holding a meeting on Google Meet to discuss and understand the topic thoroughly. This ensures that all team members are aligned and have a clear understanding of the objectives. Following the discussion, we collectively search for relevant resources to support our work and determine the overall outline of the task, including the tools and methods we will use to complete it efficiently. Once the outline is established, we divide the task into smaller, manageable parts and assign them equally among the team members. For instance, Task 1.1, which covers the ground floor, is handled by Pei Wen and Zhao Hong, while Task 1.2, focused on the first floor, is assigned to me and Koo Xuan. To ensure seamless collaboration, we create a shared workspace, such as a Google Docs document, where everyone can contribute their work in real-time. This setup not only facilitates tracking progress but also

allows for easy integration of all team members' contributions. Finally, once all parts are completed, we review the entire task together to ensure consistency and accuracy before finalizing it. This structured approach ensures effective teamwork and high-quality results.

### **Explain what you have learned from working as a group.**

Working as a group has taught me several valuable lessons. First and foremost, I learned the importance of collaboration and effective communication in achieving a common goal. Sharing responsibilities among team members allowed us to leverage each person's unique strengths, ensuring tasks were completed more efficiently. I also gained experience in resolving conflicts constructively, which helped maintain a positive and productive working environment. Additionally, the process highlighted the value of time management and mutual support, especially when meeting deadlines. Overall, working in a group has significantly enhanced my teamwork skills and prepared me for real-world collaborative projects by teaching me how to work cohesively and adapt to different perspectives.

### **Explain what you have learned from doing the project**

Working on this project has significantly enhanced my understanding of fundamental networking concepts. I gained knowledge about network topologies and how different designs impact efficiency and scalability. I also learned the practical steps for setting up connections between devices, including selecting appropriate hardware and estimating costs to create an effective and budget-friendly network environment. Furthermore, I developed an understanding of key devices, such as switches and patch panels, and their roles in a network setup. This hands-on experience allowed me to connect theoretical concepts with real-world applications.

### **Your comments and suggestions on the project.**

This project has been an enriching experience that deepened my expertise in networking. I particularly appreciated the opportunity to delve into complex topics, such as IP address assignment and subnet calculations, which are essential for configuring a functional network. The project provided a solid foundation for understanding the challenges and considerations involved in network design and implementation. However, I believe incorporating a simulation tool, such as Cisco Packet Tracer or GNS3, could enhance the learning experience. This addition would allow students to test and visualize their network setups in a virtual environment, bridging the gap between theoretical planning and practical application.